Charotar University of Science and Technology
M.Sc. (Nano Science & Technology) Semester II Examination May 2011

NT709 Fabrication of Nanomaterials-I

Date 4th May 2011 **Time: 10.00 am to 10.30 pm** **Max. Marks 20**

Part – A

Do as directed (Show your calculation within the space provided)

Que.1 Crystal in the nanometer scale has a (1)
(i) Low melting point and high lattice constant (ii) High melting point and high lattice constant
(iii) Low melting point and reduced lattice constant

Que.2 When the size of materials reaches de-Broglie's wavelength the band gap of semiconductor becomes, (1)
(i) Narrower (ii) Wider (iii) no effect

Que.3 The surface energy for {100} facet will be (show the calculation and tick the correct answer) (3)

(i) $4\varepsilon/a^2$ (ii) $(5/\sqrt{2})\,\varepsilon/a^2$ (iii) $(2\sqrt{3})\,\varepsilon/a^2$ (iv) ε/a^2

Que.4 Crystal surface with high miller indices process (1)
(i) Lower surface energy (ii) Higher surface energy (iii) Zero surface energy

Que.5 Surface energy of concave surface is : (1)
(i) Positive (ii) Negative (iii) does not depend

Que.6 Crystal forms facets structure above roughening temperature (True/false) (1)

Que.7 Calculate the van der Waals attraction between two blocks separated by a distance 4 nm and having the Hamaker's constant 3.14×10^{-20} J (3)

Que.8 When mass is transferred from flat surface to a concave surface the chemical potential (1)
(i) Increases (ii) decreases (iii) remains constant

Que.9 Oswald ripening is to combine individual particles to a bulk with solid interface to connect each other (True/False) (1)

Que.10 Crystal energy barrier becomes same as homogeneous nucleation if wetting angle is (1)

(a) > 180 degree (b) = 0.0 degree (c) = 180 degree

Que.11 In poly growth mechanism the growth rate is a function of (1)

(i) Particle size (ii) Temperature (iii) Time.

Que.12 Particles formed in aerosol techniques are in the range of (1)

(i) 1 to 20 meter (ii) 1 to 20 nm (iii) 1 to 20 micro meter

Que.13 A strong reducing agent favors the formation of (1)

(a) Larger particles (b) Smaller particles (c) No influence on size of particle

Que.14 Calculate the surface charge density for TiO2 particles in water at 300K and 400K. The pH of the solution at both temperatures is 8 and p.z.c of TiO2 particles in water is 6. The gas constant and Faraday's constant values are, 8.31 J/mol.K and 96,500 C/mol, respectively. (3)

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Part B
(Write each section on separate answer book)

Section I

Que 1. (a) What are the challenges in the fabrication and processing of nanomaterials and nanostructures? (3)

(b) Bottom-up approach promises a better chance to obtain nanostructures with less defects, more homogeneous chemical composition, and better short and long range ordering. Explain why? (2)

(c) The production of colloidal Au is well known from ages then what makes Nanotechnology to become so famous recently? (3)

(d) Define a relationship between nanometer and meter. (1)

(e) Give the full form of MBE. (1)

Que 2. (a) Define the surface energy, γ and each term of it. Explain why it is only applicable to solids with rigid structure? (4)

(b) What is surface relaxation and surface restructuring? Why it is needed? Explain with example. (4)

(c) Explain why the process of sintering has to be avoided to produce nanostructures materials? (2)

Section II

Que 3. (a) What are the mechanisms to produce the surface charge on the solid sphere in a polar solvent? Write Nernst equation of surface potential of a solid. (4)

(b) How van der Waals's interaction changes from two spheres of equal radius to an unequal radius and to parallel plates to two blocks? (2)

(c) What are the basic assumptions in DLVO theory and under which conditions one can apply this theory? (4)

Que 4. (a) Derive the energy barrier that a nucleation process must overcome in order to have homogeneous nucleation. (6)

(b) Explain the Sol-gel process for the synthesis if colloidal nanoparticles, Write its advantages. (4)

Que 5. (a) Explain the heterogeneous nucleation process on a planar solid substrate assuming growth species in the vapor phase. (6)

(b) What is kinetically controlled growth process? (4)

OR

Que 5. (a) How aerosol method differs from other methods for synthesis of nanoparticles. (5)

(b) What are the fundamentals for the synthesis of mono-dispersed nanoparticles? (5)

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